

AI Exploratory Test Report

Target Application: https://task-flow-100.lovable.app

Report Date: March 24, 2026 at 10:15

Tool Version: AI Exploratory Test Engine v2.0

Pages Analyzed: 6

Total Test Cases Generated: 54

Valid Test Cases: 54

Rejected (Hallucinated): 0

Tests Executed: 53

Passed: 52

Failed (Bugs Found): 1

2. Execution Results Summary

52 PASSED

Tests behaved correctly

1 FAILED

Bugs found in application

Bugs Found - Failed Test Cases:

Test ID	Goal	Expected (Not Met - Bug)
VT-51	Test dashboard error handling	error message

3. Application Coverage Summary

Metric	Value
Pages Crawled	6
Pages with Interactive Elements	6
Input Fields Identified	3
Buttons Identified	10
Navigation Links	30
Dropdowns/Selects	0
Test Cases Generated	54

Expected: validation failure

VT-02 | Test the contact form with special characters [PASSED]

- Page: /contactTarget: Your nameType: Negative Input Tests
1. Navigate to '/contact'

2. Enter '!@#\$\$%^&*()' into 'Your name'

3. Enter '!@#\$\$%^&*()' into 'you@example.com'

4. Click 'Send Message'

Expected: error message

VT-03 | Exceed the max limit of the search field [PASSED]

- Page: /tasksTarget: Search tasksType: Boundary Tests
1. Navigate to '/tasks'

2. Enter 'a'.repeat(1001) into 'Search tasks'

3. Click 'Filter tasks'

Expected: validation failure

VT-04 | Test the contact form with empty fields [PASSED]

- Page: /contactTarget: Your nameType: Zero/Empty Input Tests
1. Navigate to '/contact'

2. Enter " into 'Your name'

3. Enter " into 'you@example.com'

4. Click 'Send Message'

Expected: error message

VT-05 | Break the search functionality with a whitespace-only string [PASSED]

- Page: /tasksTarget: Search tasksType: General Exploratory Tests
1. Navigate to '/tasks'

2. Enter ' ' into 'Search tasks'

3. Click 'Filter tasks'

Expected: validation failure

VT-06 | Test the contact form with cross-field contamination [PASSED]

- Page: /contactTarget: Your nameType: General Exploratory Tests
1. Navigate to '/contact'

2. Enter 'example' into 'Your name'

3. Enter 'example' into 'you@example.com'

4. Enter 'example@example' into 'you@example.com'

5. Click 'Send Message'

Expected: error message

VT-07 | Exceed the min limit of the search field [PASSED]

- Page: /tasksTarget: Search tasksType: Boundary Tests
1. Navigate to '/tasks'

2. Enter '-1' into 'Search tasks'

3. Click 'Filter tasks'

Expected: validation failure

VT-08 | Test the search functionality with a dangerous string

Page: /tasks

Target: Search tasks

Type: Negative Input Tests

1. Navigate to '/tasks'
2. Enter '<script>alert(1)</script>' into 'Search tasks'
3. Click 'Filter tasks'

Expected: validation failure

VT-09 | Break the contact form with a long string [PASSED]

Page: /contact

Target: Your name

Type: Boundary Tests

1. Navigate to '/contact'
2. Enter '99999999999999999999999999999999' into 'Your name'
3. Enter '99999999999999999999999999999999' into 'you@example.com'
4. Click 'Send Message'

Expected: error message

VT-10 | Test the search functionality with a paste-style long string [PASSED]

Page: /tasks

Target: Search tasks

Type: Boundary Tests

1. Navigate to '/tasks'
2. Enter '0.000001' into 'Search tasks'
3. Click 'Filter tasks'

Expected: validation failure

VT-11 | Submit contact form with empty name field [PASSED]

Page: /contact

Target: Your name

Type: Validation Enforcement Tests

1. Navigate to '/contact'
2. Enter " into 'Your name'
3. Enter 'you@example.com' into 'you@example.com'
4. Click 'Send Message'

Expected: Validation failure for missing name

VT-12 | Submit contact form with empty email field [PASSED]

Page: /contact

Target: Your name

Type: Validation Enforcement Tests

1. Navigate to '/contact'
2. Enter 'John Doe' into 'Your name'
3. Enter " into 'you@example.com'
4. Click 'Send Message'

Expected: Validation failure for missing email

VT-13 | Submit contact form with valid data and then submit again [PASSED]

Page: /contact

Target: Your name

Type: General Exploratory Tests

1. Navigate to '/contact'
2. Enter 'John Doe' into 'Your name'
3. Enter 'you@example.com' into 'you@example.com'
4. Click 'Send Message'
5. Click 'Send Message'

Expected: Validation failure or error message for duplicate submission

VT-14 | Submit task creation form with empty search field [PASSED]

Page: /tasks

Target: Search tasks

Type: Validation Enforcement Tests

1. Navigate to '/tasks'
2. Enter '' into 'Search tasks'
3. Click 'Create Task'

*Expected: Validation failure for missing search query***VT-15 | Submit task creation form with valid search query and then s [PASSED]**

Page: /tasks

Target: Search tasks

Type: General Exploratory Tests

1. Navigate to '/tasks'
2. Enter 'test task' into 'Search tasks'
3. Click 'Create Task'
4. Click 'Create Task'

*Expected: Validation failure or error message for duplicate submission***VT-16 | Submit contact form with partially filled fields [PASSED]**

Page: /contact

Target: Your name

Type: General Exploratory Tests

1. Navigate to '/contact'
2. Enter 'John Doe' into 'Your name'
3. Click 'Send Message'

*Expected: Validation failure for missing email***VT-17 | Submit task creation form with partially filled fields [PASSED]**

Page: /tasks

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/tasks'
2. Click 'Create Task'

*Expected: Validation failure for missing search query***VT-18 | Submit contact form with valid data, see result, and then su [PASSED]**

Page: /contact

Target: Your name

Type: General Exploratory Tests

1. Navigate to '/contact'
2. Enter 'John Doe' into 'Your name'
3. Enter 'you@example.com' into 'you@example.com'
4. Click 'Send Message'
5. Enter 'Jane Doe' into 'Your name'
6. Click 'Send Message'

*Expected: Validation failure or error message for modified submission***VT-19 | Submit task creation form with valid search query, see resul [PASSED]**

Page: /tasks

Target: Search tasks

Type: General Exploratory Tests

1. Navigate to '/tasks'
2. Enter 'test task' into 'Search tasks'
3. Click 'Create Task'
4. Enter 'new task' into 'Search tasks'
5. Click 'Create Task'

Expected: Validation failure or error message for modified submission

VT-20 | Submit contact form with valid data and then dismiss the for [PASSED]

Page: /contact

Target: Your name

Type: General Exploratory Tests

1. Navigate to '/contact'
2. Enter 'John Doe' into 'Your name'
3. Enter 'you@example.com' into 'you@example.com'
4. Click 'Send Message'
5. Click 'Dismiss'

*Expected: Error message for dismissed form***VT-21 | Break the Search tasks input with a very large number [PASSED]**

Page: /tasks

Target: Search tasks

Type: Boundary Tests

1. Navigate to '/tasks'
2. Enter '9999999999' into 'Search tasks'
3. Click 'Filter tasks'

*Expected: Invalid input error***VT-22 | Test the Your name input with a very small decimal [PASSED]**

Page: /contact

Target: Your name

Type: Boundary Tests

1. Navigate to '/contact'
2. Enter '0.0001' into 'Your name'
3. Click 'Send Message'

*Expected: Validation failure***VT-23 | Probe the you@example.com input with a zero value [PASSED]**

Page: /contact

Target: you@example.com

Type: Boundary Tests

1. Navigate to '/contact'
2. Enter '0' into 'you@example.com'
3. Click 'Send Message'

*Expected: Error message***VT-24 | Test the Your name input with one above maximum value [PASSED]**

Page: /contact

Target: Your name

Type: General Exploratory Tests

1. Navigate to '/contact'
2. Enter 'a' * 1000 into 'Your name'
3. Click 'Send Message'

*Expected: Validation failure***VT-25 | Probe the you@example.com input with a very large number [PASSED]**

Page: /contact

Target: you@example.com

Type: Boundary Tests

1. Navigate to '/contact'
2. Enter '9999999999' into 'you@example.com'
3. Click 'Send Message'

Expected: Error message

VT-26 | Break the Search tasks input with exact minimum allowed valu [PASSED]

Page: /tasks

Target: Search tasks

Type: Boundary Tests

1. Navigate to '/tasks'
2. Enter '0' into 'Search tasks'
3. Click 'Filter tasks'

Expected: Invalid input error

VT-27 | Test the Your name input with exact maximum allowed value [PASSED]

Page: /contact

Target: Your name

Type: General Exploratory Tests

1. Navigate to '/contact'
2. Enter 'a' * 50 into 'Your name'
3. Click 'Send Message'

Expected: Validation failure

VT-28 | Probe the you@example.com input with a very small decimal [PASSED]

Page: /contact

Target: you@example.com

Type: Boundary Tests

1. Navigate to '/contact'
2. Enter '0.0001' into 'you@example.com'
3. Click 'Send Message'

Expected: Error message

VT-29 | Break form submission by rapid input [PASSED]

Page: /contact

Target: Your name

Type: General Exploratory Tests

1. Navigate to '/contact'
2. Enter 'John Doe' into 'Your name'
3. Enter 'john@example.com' into 'you@example.com'
4. Click 'Send Message'
5. Enter 'Jane Doe' into 'Your name'
6. Enter 'jane@example.com' into 'you@example.com'
7. Click 'Send Message'

Expected: Validation failure for duplicate submission

VT-30 | Test multi-step form interaction with clear button [PASSED]

Page: /contact

Target: Your name

Type: General Exploratory Tests

1. Navigate to '/contact'
2. Enter 'John Doe' into 'Your name'
3. Enter 'john@example.com' into 'you@example.com'
4. Click 'Dismiss'
5. Enter 'Jane Doe' into 'Your name'
6. Enter 'jane@example.com' into 'you@example.com'
7. Click 'Send Message'

Expected: Error message for incomplete form

VT-31 | Break search functionality with rapid input [PASSED]

Page: /tasks	Target: Search tasks	Type: General Exploratory Tests
1. Navigate to '/tasks' 2. Enter 'task1' into 'Search tasks' 3. Click 'Filter tasks' 4. Enter 'task2' into 'Search tasks' 5. Click 'Filter tasks' 6. Enter 'task3' into 'Search tasks' 7. Click 'Filter tasks'		

Expected: No UI errors or crashes

VT-32 | Break form submission with multiple clicks [PASSED]

Page: /contact	Target: Your name	Type: General Exploratory Tests
1. Navigate to '/contact' 2. Enter 'John Doe' into 'Your name' 3. Enter 'john@example.com' into 'you@example.com' 4. Click 'Send Message' 5. Click 'Send Message' 6. Click 'Send Message'		

Expected: Error message for duplicate submission

VT-33 | Test search functionality with no input [PASSED]

Page: /tasks	Target: N/A	Type: General Exploratory Tests
1. Navigate to '/tasks' 2. Click 'Filter tasks'		

Expected: No results found

VT-34 | Test form submission with all fields filled [PASSED]

Page: /contact	Target: Your name	Type: General Exploratory Tests
1. Navigate to '/contact' 2. Enter 'John Doe' into 'Your name' 3. Enter 'john@example.com' into 'you@example.com' 4. Click 'Send Message'		

Expected: Validation failure for unknown reason

VT-35 | Break search functionality with special characters [PASSED]

Page: /tasks	Target: Search tasks	Type: Negative Input Tests
1. Navigate to '/tasks' 2. Enter '!@#\$\$' into 'Search tasks' 3. Click 'Filter tasks'		

Expected: No UI errors or crashes

VT-36 | Break mode switching on tasks page [PASSED]

Page: /tasks

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/tasks'
2. Click 'Filter tasks'
3. Click 'Create Task'
4. Click 'Filter tasks'

*Expected: No UI errors***VT-37 | Rapidly apply and unapply mode on tasks page [PASSED]**

Page: /tasks

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/tasks'
2. Click 'Filter tasks'
3. Click 'Create Task'
4. Click 'Filter tasks'
5. Click 'Create Task'

*Expected: No UI errors***VT-38 | Click mode button multiple times on tasks page [PASSED]**

Page: /tasks

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/tasks'
2. Click 'Filter tasks'
3. Click 'Filter tasks'
4. Click 'Filter tasks'
5. Click 'Create Task'

*Expected: Validation failure***VT-39 | Submit with new labels after mode switch on tasks page [PASSED]**

Page: /tasks

Target: Search tasks

Type: General Exploratory Tests

1. Navigate to '/tasks'
2. Click 'Filter tasks'
3. Enter 'test' into 'Search tasks'
4. Click 'Create Task'

*Expected: Error message***VT-40 | Switch between modes on tasks page with incomplete fields [PASSED]**

Page: /tasks

Target: Search tasks

Type: Navigation & Cross-Page Tests

1. Navigate to '/tasks'
2. Click 'Filter tasks'
3. Click 'Create Task'
4. Enter 'test' into 'Search tasks'
5. Click 'Create Task'

Expected: Validation failure

VT-41 | Test mode switching with rapid navigation on tasks page [PASSED]

Page: /tasks

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/tasks'
2. Click 'Filter tasks'
3. Navigate to '/tasks'
4. Click 'Create Task'
5. Click 'Filter tasks'

*Expected: No UI errors***VT-42 | Break mode switching with multiple mode switches on tasks pa [PASSED]**

Page: /tasks

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/tasks'
2. Click 'Filter tasks'
3. Click 'Create Task'
4. Click 'Filter tasks'
5. Click 'Create Task'
6. Click 'Filter tasks'

*Expected: Error message***VT-43 | Test mode switching with search on tasks page [PASSED]**

Page: /tasks

Target: Search tasks

Type: General Exploratory Tests

1. Navigate to '/tasks'
2. Enter 'test' into 'Search tasks'
3. Click 'Filter tasks'
4. Click 'Create Task'

*Expected: No UI errors***VT-44 | Rapidly switch modes and submit on tasks page [PASSED]**

Page: /tasks

Target: Search tasks

Type: General Exploratory Tests

1. Navigate to '/tasks'
2. Click 'Filter tasks'
3. Click 'Create Task'
4. Click 'Filter tasks'
5. Click 'Create Task'
6. Enter 'test' into 'Search tasks'

*Expected: Validation failure***VT-45 | Break mode switching with multiple rapid mode switches on ta [PASSED]**

Page: /tasks

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/tasks'
2. Click 'Filter tasks'
3. Click 'Create Task'
4. Click 'Filter tasks'
5. Click 'Create Task'
6. Click 'Filter tasks'
7. Click 'Create Task'

Expected: Error message

VT-46 | Trigger error on contact form submission [PASSED]

Page: /contact

Target: N/A

Type: Validation Enforcement Tests

- 1. Navigate to '/contact'
- 2. Click 'Send Message'

Expected: error message

VT-47 | Recover from invalid task search [PASSED]

Page: /tasks

Target: Search tasks

Type: Negative Input Tests

- 1. Navigate to '/tasks'
- 2. Enter 'invalid' into 'Search tasks'
- 3. Click 'Filter tasks'
- 4. Enter 'valid' into 'Search tasks'
- 5. Click 'Filter tasks'

Expected: error clearance

VT-48 | Test error handling on task creation [PASSED]

Page: /tasks

Target: N/A

Type: General Exploratory Tests

- 1. Navigate to '/tasks'
- 2. Click 'Create Task'
- 3. Click 'Create Task' again

Expected: error message

VT-49 | Clear contact form after submission [PASSED]

Page: /contact

Target: Your name

Type: General Exploratory Tests

- 1. Navigate to '/contact'
- 2. Enter 'name' into 'Your name'
- 3. Enter 'email' into 'you@example.com'
- 4. Click 'Send Message'
- 5. Click 'Dismiss'
- 6. Click 'Send Message'

Expected: error message

VT-50 | Recover from empty task search [PASSED]

Page: /tasks

Target: Search tasks

Type: Zero/Empty Input Tests

- 1. Navigate to '/tasks'
- 2. Click 'Filter tasks'
- 3. Enter 'search' into 'Search tasks'
- 4. Click 'Filter tasks'

Expected: error clearance

VT-51 | Test dashboard error handling [FAILED - BUG FOUND]

Page: /dashboard

Target: N/A

Type: General Exploratory Tests

- 1. Navigate to '/dashboard'
- 2. Click 'Dismiss'
- 3. Click 'Dismiss' again

Expected: error message

!! Application did not meet expected behavior - potential bug detected.

VT-52 | Submit contact form with invalid email [PASSED]

Page: /contactTarget: Your nameType: Negative Input Tests

1. Navigate to '/contact'

2. Enter 'name' into 'Your name'

3. Enter 'invalid' into 'you@example.com'

4. Click 'Send Message'

5. Enter 'email' into 'you@example.com'

6. Click 'Send Message'

Expected: error clearance

VT-53 | Test task creation with invalid input [PASSED]

Page: /tasksTarget: Search tasksType: Negative Input Tests

1. Navigate to '/tasks'

2. Click 'Create Task'

3. Enter 'invalid' into 'Search tasks'

4. Click 'Create Task'

5. Enter 'valid' into 'Search tasks'

6. Click 'Create Task'

Expected: error clearance

VT-54 | Recover from contact form error [PASSED]

Page: /contactTarget: Your nameType: Validation Enforcement Tests

1. Navigate to '/contact'

2. Enter 'name' into 'Your name'

3. Click 'Send Message'

4. Enter 'email' into 'you@example.com'

5. Click 'Send Message'

Expected: error clearance

7. Structural Validation Confirmation

All 54 test cases were validated against the UI exploration snapshot. The validation engine verified that:

- ✓

All referenced UI element labels exist in the application snapshot.
- ✓

All referenced page paths exist in the crawled page inventory.
- ✓

No hallucinated or invented element labels were detected.
- ✓

Each test case has a valid structure: ID, goal, steps (min 2), and expected result.

8. Appendix

Snapshot Statistics

Metric	Value
Snapshot File	ai_exploration_snapshot.json

Snapshot Size	11575 bytes
Total Pages in Snapshot	6
Total Input Fields	3
Total Buttons	10
Total Navigation Links	30

Pipeline Configuration

Parameter	Value
AI Model	llama-3.3-70b-versatile (Groq)
Generation Batches	6
Tests per Batch	~10
Batch Delay	30 seconds
Validation Engine	3-Layer Anti-Hallucination
Step Generator	Deterministic (Template-Based)

Output Files Reference

File	Path
Raw Test Cases	data/generated_test_cases.json
Validated Test Cases	data/validated_test_cases.json
Rejected Test Cases	data/rejected_test_cases.json
Gauge Specification	specs/ai_exploration.spec
Step Implementations	step_impl/step_implementation.py
This Report	reports/test_report.pdf